Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $x > 0$, and $\log_2 x + \log_2(\sqrt{x}) + \log_2(\sqrt[4]{x}) + \log_2(\sqrt[8]{x}) + \log_2(\sqrt[16]{x}) + \ldots = 4$ then x equals

(A) 2

(B) 3

(C) 4

(D) 5

Q.2

For $0 < \phi < \pi/2$ if $x = \sum_{n=0}^{\infty} \cos^{2n} \phi, y = \sum_{n=0}^{\infty} \sin^{2n} \phi, z = \sum_{n=0}^{\infty} \cos^{2n} \phi \sin^{2n} \phi$, then

(A) $xyz = xz + y$

(B) $xyz = xy + z$

(C) $xyz = x + y + z$

(D) $xyz = yz + x$

Q.3

If the sum of the series $2 + \frac{5}{x} + \frac{25}{x^2} + \frac{125}{x^3} + \ldots$ is finite, then

(A) $|x| > 5$

(B) $-5 < x < 5$

(C) $|x| < 5/2$

(D) $|x| > 5/2$

Q.4

Let A_n be the sum of the first n terms of the geometric series $704 + \frac{704}{2} + \frac{704}{4} + \frac{704}{8} + \ldots$
and B_n be the sum of the first n terms of the geometric series $1984 - \frac{1984}{2} + \frac{1984}{4} - \frac{1984}{8} + \ldots$
If $A_n = B_n$, then the value of n is (where $n \in N$).

(A) 4

(B) 5

(C) 6

(D) 7

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Q.5

If $|r| > 1$ and $x = a + \frac{a}{r} + \frac{a}{r^2} + \ldots$ to ∞,
$y = b - \frac{b}{r} + \frac{b}{r^2} - \ldots$..to ∞
and $z = c + \frac{c}{r^2} + \frac{c}{r^4} + \ldots$ to ∞, then $\frac{xy}{z} =$

(A) $\frac{ab}{c}$

(B) $\frac{ac}{b}$

(C) $\underline{bc}$

(D) 2

Q.6

a, b, c are in G.P. with $1 < a < b < n$, and $n > 1$ is an integer. $\log_a n, \log_b n, \log_c n$ form a sequence. This sequence is which one of the following ?

(A) Harmonic progression

(B) Arithmetic progression

(C) Geometric progression

(D) None of these

Q.7

If S, P and R are the sum, product and sum of the reciprocals of n terms of an increasing G.P respectively and $S^n = R^n \cdot P^k$, then k is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Q.8

A G.P. consists of an even number of terms. If the sum of all the terms is 5 times the sum of terms occupying odd places, then the common ratio is

(A) 5

(B) 1

(C) 4

(D) 3

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Q.9

The value of x in $(0, \pi)$ which satisfy the equation $8^{1+|\cos x|+\cos^2 x+|\cos^3 x|+\ldots\ldots\ldots \text{ to } \infty} = 4^3$ is

(A) $\left\{\frac{\pi}{2}, \frac{3\pi}{4}\right\}$

(B) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$

(C) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$

(D) $\left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$

Q.10

If $a_1, a_2 \ldots\ldots, a_n$ are positive real numbers whose
product is a fixed number c, then the minimum value of $a_1 + a_2 + \ldots\ldots + a_{n-1} + 2a_n$ is

(A) $n(2c)^{1/n}$

(B) $(n+1)c^{1/n}$

(C) $2nc^{1/n}$

(D) $(n+1)(2c)^{1/n}$

Q.11

If a, b, c are in G.P., then

(A) $a^2,\ b^2, c^2$ are in G.P.

(B) $a^2(b+c), c^2(a+b), b^2(a+c)$ are in G.P.

(C) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in G.P.

(D) None of these

Q.12

Three numbers are in G.P. such that their sum is 38 and their product is 1728 . The greatest number among them is:

(A) 18

(B) 16

(C) 14

(D) None of these

Q.13

There are four numbers of which the first three are in G.P. and the last three are in A.P., whose common difference is 6 . If the first and the last numbers are equal then two other numbers are

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(A) $-2, 4$

(B) $-4, 2$

(C) 2,6

(D) none

Q.14

The sum of an infinite GP is x and the common ratio r is such that $|r| < 1$. If the first term of the GP is 2, then which one of the following is correct?

(A) $-1 < x < 1$

(B) $-\infty < x < 1$

(C) $1 < x < \infty$

(D) None of these

Q.15

If $x = 1 + a + a^2 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots$..to infinity and $y = 1 + b + b^2 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots$ infinity, where a, b are proper fractions, then $1 + ab + a^2b^2 + \ldots$..to infinity is equal :

(A) $\frac{xy}{x+y-1}$

(B) $\frac{xy}{x-y-1}$

(C) $\frac{xy}{x-y+1}$

(D) $\frac{xy}{x+y+1}$

Q.16

A series is such that its every even term is 'a 'times the term before it and every odd term is c times the term before it. The sum of $2n$ term of the series is (the first term is unity)

(A) $\frac{(1-c^n)(1-a^n)}{1-ac}$

(B) $\frac{(1+a)(1-c^n a^n)}{1-ac}$

(C) $\frac{(1+c^n)(1+a^n)}{1-ac}$

(D) $\frac{(1+a)(1+c^n a^n)}{1+ac}$

Q.17

Let $a = 111\ldots1(55$ digits $)$
$b = 1 + 10 + 10^2 + \ldots + 10^4$
$c = 1 + 10^5 + 10^{10} + 10^{15} + \ldots + 10^{50}$, then

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(A) $a = b + c$

(B) $a = bc$

(C) $b = ac$

(D) $c = ab$

Q.18

Which one of the following options is correct?

(A) $\sin^2 30^\circ, \sin^2 45^\circ, \sin^2 60^\circ$ are in GP

(B) $\cos^2 30^\circ, \cos^2 45^\circ, \cos^2 60^\circ$ are in GP

(C) $\cot^2 30^\circ, \cot^2 45^\circ, \cot^2 100^\circ$ are in GP

(D) $\tan^2 30^\circ, \tan^2 45^\circ, \tan^2 60^\circ$ are in GP

Q.19

The 100th term of the sequence $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, \ldots$ is

(A) 12

(B) 13

(C) 14

(D) 15

Q.20

What is the product of first $2n + 1$ terms of a geometric progression ?

(A) The $(n + 1)$ th power of the nth term of the GP

(B) The $(2n + 1)$ th power of the nth term of the GP

(C) The $(2n + 1)$ th power of the $(n + 1)$ th term of the GP

(D) The nth power of the $(n + 1)$ th terms of the GP

Answers & Solutions

Q.1 Answer:

4

Solution:

We can write the given equation as

$$\log_2 \left(x^{1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots}\right) = 4$$

$$\Rightarrow \quad \log_2 \left(x^2\right) = 4 \quad \Rightarrow \quad x^2 = 2^4 \Rightarrow x = 4$$

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Q.2 Answer:

$xyz = xy + z$

Solution:

Given $x = \sum_{n=0}^{\infty} \cos^{2n}\phi, \quad y = \sum_{n=0}^{\infty} \sin^{2n}\phi$ **and** $z = \sum_{n=0}^{\infty} \cos^{2n}\phi \sin^{2n}\phi$

since $0 < \phi < \frac{\pi}{2}$**, so each series is geometric series with common ratio** $r < 1$**.**

Therefore, the series are convergent.

Now, $x = \frac{1}{1-\cos^2\phi}$

$= \frac{1}{\sin^2\phi} \quad \left(\because S_\infty = \frac{a}{1-r}\right)$

$y = \frac{1}{1-\sin^2\phi} \quad \left(\because S_\infty = \frac{a}{1-r}\right)$

$= \frac{1}{\cos^2\phi}$

$z = \frac{1}{1-\sin^2\phi\cos^2\phi}$

$\left(\because S_\infty = \frac{a}{1-r}\right)$

Consider, $xyz = \frac{1}{\sin^2\phi\cos^2\phi(1-\sin^2\phi\cos^2\phi)}$

$\text{Also, } = \frac{1}{\sin^2\phi\cos^2\phi} + \frac{1}{1-\sin^2\phi\cos^2\phi}$

$xy + z = \frac{1-\sin^2\phi\cos^2\phi+\sin^2\phi\cos^2\phi}{\sin^2\phi\cos^2\phi(1-\sin^2\phi\cos^2\phi)}$

$= \frac{1}{\sin^2\phi\cos^2\phi(1-\sin^2\phi\cos^2\phi)}$

$= xyz \quad [From(1)]$

Q.3 Answer:

$|x| > 5$

Solution:

We can rewrite the series as

$1 + 1 + \frac{5}{x} + \left(\frac{5}{x}\right)^2 + \left(\frac{5}{x}\right)^3 + \ldots$

We can sum up this series if $\left|\frac{5}{x}\right| < 1 \Leftrightarrow |x| > 5$

Q.4 Answer:

5

Solution:

$A_n = 704 + \frac{704}{2} + \frac{704}{4} + \ldots \text{ to } n \text{ terms}$

$= \frac{704\left(1-\left(\frac{1}{2}\right)^n\right)}{1-\frac{1}{2}} = 704 \times 2\left(1 - \left(\frac{1}{2}\right)^n\right)$

$B_n = 1984 - \frac{1984}{2} + \frac{1984}{4} \ldots \text{ to } n \text{ terms}$

$= \frac{1984\left(1-\left(\frac{-1}{2}\right)^n\right)}{1-\left(\frac{-1}{2}\right)} = 1984 \times \frac{2}{3}\left(1 - \left(\frac{-1}{2}\right)^n\right)$

$\text{Now, } A_n = B_n \Rightarrow 704 \times 2\left(1 - \left(\frac{1}{2}\right)^n\right)$

$= 1984 \times \frac{2}{3} \times \left(1 - \left(\frac{-1}{2}\right)^n\right)$

$\Rightarrow \quad 33 - 31 = 33\left(\frac{1}{2}\right)^n - 31\left(\frac{-1}{2}\right)^n$

$\Rightarrow 2^{n+1} = 33 - 31(-1)^n \quad \Rightarrow \quad n = 5$

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Q.5 Answer:

$\frac{ab}{c}$

Solution:

Since $|r| > 1$, $\frac{1}{|r|} < 1$

$\therefore \quad x = \frac{a}{1-\frac{1}{r}} = \frac{ar}{r-1}$

Similarly, $y = \frac{b}{1-\left(-\frac{1}{r}\right)} = \frac{br}{r+1}$ **and**

$z = \frac{c}{1-\frac{1}{r^2}} = \frac{cr^2}{r^2-1}$ **...(1)**

$\therefore \quad xy = \frac{ar}{r-1} \times \frac{br}{r+1} = \frac{abr^2}{r^2-1}$ **...(2)**

Dividing (2) **by** (1)**, we get**

$\frac{xy}{z} = \frac{abr^2}{r^2-1} \times \frac{r^2-1}{cr^2} = \frac{ab}{c}$

Q.6 Answer:

Harmonic progression

Solution:

If a, b, c **are in G.P. then,** $b^2 = ac \Rightarrow b = (ac)^{1/2}$ $\ldots..(1)$

Taking $\log_n$ **on both the sides of eq. (1).** $\log_n b = \frac{1}{2}\left[(\log_n(ac)\right] = \frac{\log_n a + \log_n c}{2}$

or, $\frac{\log_n a + \log_n c}{2} = \log_n b$

So, $\log_n a, \log_n b$ **and** $\log_n c$ **are in AP.**

Hence, $\frac{1}{\log_n a}, \frac{1}{\log_n b}, \frac{1}{\log_n c}$ **are in H.P**

$\log_a n = \frac{1}{\log_n a}$

$\log_b n = \frac{1}{\log_n b}$

$\log_c n = \frac{1}{\log_n c}$

i.e. $\log_a n, \log_h n,$ **and** $\log_c n$ **are in HP.**

Q.7 Answer:

2

Solution:

$S = \frac{a(1-r^n)}{1-r}, P = a^n \cdot r^{\frac{n(n-1)}{2}}$

$R = \frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \ldots\ldots n$ **terms** $= \frac{1-r^n}{a(1-r)r^{n-1}}$

$S^n = R^n P^k \Rightarrow \left(\frac{S}{R}\right)^n = P^k$

$\Rightarrow \left(a^2 r^{n-1}\right)^n = P^k \Rightarrow P^2 = P^k \Rightarrow k = 2$

Q.8 Answer:

4

Solution:

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Let the G.P. be $a, ar, ar^2, \ldots\ldots\ldots$

$S = a + ar + ar^2 + \ldots\ldots\ldots +$ **to $2n$ term** $= \frac{a\left(r^{2n}-1\right)}{r-1}$

The series formed by taking term occupying odd places is $S_1 = a + ar^2 + ar^4 + \ldots\ldots\ldots\ldots$ **to n terms** $S_1 = \frac{a\left[\left(r^2\right)^n-1\right]}{r^2-1} \Rightarrow S_1 = \frac{a\left(r^{2n}-1\right)}{r^2-1}$

Now, $S = 5S_1$ **or** $\frac{a\left(r^{2n}-1\right)}{r-1} = 5\frac{a\left(r^{2n}-1\right)}{r^2-1}$

$\Rightarrow \quad 1 = \frac{5}{r+1} \Rightarrow r + 1 = 5 \therefore r = 4$

Q.9 Answer:

$\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$

Solution:

We have $8^{1+|\cos x|+|\cos x|^2+|\cos x|^3+\ldots\ldots\ldots \text{ to } \infty} = 4^3$

$\left[\because \cos^2 x = \left|\cos^2 x\right| \text{ also } \left|\cos^n x\right| = |\cos x|^n\right]$

$\Rightarrow 8^{\frac{1}{1-|\cos x|}} = 4^3 \Rightarrow 2^{\frac{3}{1-|\cos x|}} = 2^6$

$\therefore \frac{3}{1-|\cos \mathrm{x}|} = 6 \Rightarrow 1 - |\cos \mathrm{x}| = \frac{1}{2}$

$\therefore |\cos x| = \frac{1}{2} \Rightarrow \cos x = \pm\frac{1}{2}$

$\therefore \mathrm{x} = \frac{\pi}{3}, \frac{2\pi}{3}$

Q.10 Answer:

$\mathrm{n}(2\mathrm{c})^{1/\mathrm{n}}$

Solution:

We have $\left(a_1 + a_2 + \ldots\ldots + a_{n-1} + 2a_n\right)/n \geq \left(a_1 a_2 \cdots .. a_{n-1} 2a_n\right)^{1/n}$

$\Rightarrow a_1 + a_2 + a_3 + \ldots. + a_{n-1} + 2a_n \geq n(2c)^{1/n}$

Q.11 Answer:

$\mathrm{a}^2,\ \mathrm{b}^2, \mathrm{c}^2$ are in G.P.

Solution:

$\because a, b, c$ **are in G.P.**

$\therefore \frac{b}{a} = \frac{c}{b} = r \Rightarrow \frac{b^2}{a^2} = \frac{c^2}{b^2} = r^2$

$\Rightarrow a^2, b^2, c^2$ **are in G.P**

Q.12 Answer:

18

Solution:

Let the required three numbers of G.P. be

$\frac{a}{r}, a$ **and ar.**

Then, their sum $= \frac{a}{r} + a + ar = 38$

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$\Rightarrow a\left(\frac{1+r+r^2}{r}\right)=38$

product $=\frac{a}{r}\times a\times ar=1728$

$\Rightarrow \quad a^3=(12)^3 \quad \therefore \quad a=12$

Substitute the value of a**, in equation (i), we get** $12\times\left(\frac{1+r+r^2}{r}\right)=38$

$\Rightarrow \quad 6+6r+6r^2=19r\Rightarrow 6r^2-13r+6=0$

$\Rightarrow \quad (3r-2)(2r-3)=0 \quad \therefore r=\frac{2}{3}$ **or** $\frac{3}{2}$

Hence, the required numbers are $18,12,8$ **or 8 , 12,18**

Greatest number $=18$

Q.13 Answer:

$-4,2$

Solution:

Let the last three numbers in A.P. be a, a $+6, a+12$**, then the first term is also** $a+12$**. But** $\mathrm{a}+12, \mathrm{a}, \mathrm{a}+6$ **are in G.P.** $\therefore a^2=(a+12)(a+6)\Rightarrow a^2=a^2+18a+72$

$\therefore \mathrm{a}=-4$

$\therefore$ **The numbers are** $8,-4,2,8$.

Q.14 Answer:

$1<x<\infty$

Solution:

$\mathrm{GP}=\mathrm{x}$

$\frac{a}{1-r}=x$ **(where,** $a=$ **lst term and** $r=$ **common ratio)**

$\Rightarrow \frac{2}{1-r}=x \quad \ldots(\mathrm{i})(\because$ **Given** $\mathrm{a}=2$ **and** $|\mathrm{r}|<1)$

$\Rightarrow -1<\mathrm{r}<1\Rightarrow 1>-\mathrm{r}>-1$

$\Rightarrow 1+1>1-r>1-1$

$\Rightarrow 0<1-r<2$

$\Rightarrow \frac{1}{1-\mathrm{r}}>\frac{1}{2}, \frac{2}{1-\mathrm{r}}>1$

from equation (i) $x>1$ **Hence,** $1<x<\infty$

Q.15 Answer:

$\frac{xy}{x+y-1}$

Solution:

If $a, ar, ar^2, ar^3 \ldots\ldots\ldots$ **are in G.P., then**

sum of infinite G.P. $=a+ar+\ldots..+\infty=\frac{a}{1-r}$ **where** $'a'$ **is the first term and** $' r'$ **is the common ratio of G.P.**

Given $x=1+a+a^2+\ldots\ldots\infty$

This is a GP, with common ratio ' a **'.**

$\Rightarrow x=\frac{1}{1-a}\Rightarrow x-ax=1\Rightarrow a=\frac{x-1}{x}$

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Again, $y = 1 + b + b^2 + \ldots\ldots\infty$ **This is also a G.P., with common ratio '** b **'.**

$\Rightarrow y = \frac{1}{1-b} \Rightarrow b = \frac{y-1}{y}$

Now, consider $1 + ab + a^2b^2 + \ldots\ldots\infty$

which is again a **GP with common ratio '** ab'**.**

$$\therefore \text{ Sum } = \frac{1}{1-ab} = \frac{1}{1 - \frac{x-1}{x} \cdot \frac{y-1}{y}}$$

$$= \frac{xy}{xy - xy + x + y - 1} = \frac{xy}{x + y - 1}$$

Q.16 Answer:

$\frac{(1+a)(1-c^n a^n)}{1-ac}$

Solution:

The required series is

$1 + a + ac + a(ac) + c(aca) + a(caca) + \ldots.$ to $2n$ terms

$= 1 + a + ac + a^2c + a^2c^2 + c^2a^3 + \ldots.$ to $2n$ terms

$= \left(1 + ac + a^2c^2 + \ldots \text{ to } n \text{ terms }\right) + \left(a + a^2c + a^3c^2 + \ldots \text{ to } n \text{ terms }\right)$ ..(i)

Sum on n **terms of G.P** $= a\left(1 - r^n\right)/(1 - r)$

So equation (i) becomes

$= 1\left(1 - (ac)^n\right)/(1 - ac) + a\left(1 - (ac)^n\right)/(1 - ac)$

$= (1 - (ac)^n(1 + a)/(1 - ac)$

$= (1 + a)\left(1 - c^n a^n\right)/(1 - ac)$

Q.17 Answer:

$a = bc$

Solution:

$a = 1 + 10 + 10^2 + \ldots + 10^{54}$

$$= \frac{10^{55} - 1}{10 - 1} = \frac{10^{55} - 1}{10^5 - 1} \cdot \frac{10^5 - 1}{10 - 1} = bc$$

Q.18 Answer:

$\tan^2 30^\circ, \tan^2 45^\circ, \tan^2 60^\circ$ are in GP

Solution:

Three numbers a, b and c will be in GP. if $b^2 = ac$**. Only option (d) i.e.** $\tan^2 30^\circ, \tan^2 45^\circ$ **and** $\tan^2 60^\circ$

are in G.P.

$\tan^2 30^\circ = \frac{1}{3}$

$\tan^2 45^\circ = 1$

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and $\tan^2 60^\circ = 3$

$\therefore \tan^2 30^\circ, \tan^2 45^\circ$ **and** $\tan^2 60^\circ$ **are in G.P.**

Q.19 Answer:

14

Solution:

1st **term** $\rightarrow 1, 2^{\text{nd}}$ **term** $= 2, 4^{\text{th}}$ **term** $\rightarrow 3, 7^{\text{th}}$ **term** $\rightarrow 4, 11^{\text{th}}$ **term** $\rightarrow 5, \ldots$

Series is $1, 2, 4, 7, 11, \ldots$

$a_n = 1 + \frac{n(n-1)}{2} = \frac{n^2-n+2}{2}$

If $n = 14$**, then** $a_n = 92$**, If** $n = 15$**, then** $a_n = 106$

Q.20 Answer:

The $(2n+1)$ th power of the $(n+1)$ th term of the GP

Solution:

The GP is a, ar, ar $^2, \ldots\ldots\ldots ar^{2n}$

So, $\mathrm{P} = \mathrm{a} \cdot \mathrm{ar} \cdot \mathrm{ar}^2 \cdot \mathrm{ar}^3 \ldots\ldots\ldots\ldots ar^{2n}$

$= \mathrm{a}^{2n+1} \cdot \mathrm{r}^{1+2+\ldots\ldots+2n}$

$= a^{(2n+1)} r^{\frac{2n(2n+1)}{2}} = a^{2n+1} r^{n(2n+1)} = (ar^n)^{(2n+1)}$

$= (2n+1)$ **th power of the** $(n+1)$ **th term of G.P.**

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